

Results and Discussion

mypy-as-oracle proof-of-detection I

mypy-as-oracle proof-of-detection II

tests/test_mypy_oracle.py runs mypy --strict as a real subprocess against every fixture under tests/mypy_fixtures/ — **six** known-bad programs, one per ISC-2/4/15/18/31/32 — and asserts a **non-zero** exit code plus the presence of a genuine error: line on each (ISC-19, reusing a consumed protocol-phase instance, deliberately has **no** fixture here: it is a runtime-only discipline, not a static one — see @sec:type-architecture — so there is nothing for a static oracle to catch). **Three** known-good fixtures are the suite's positive controls, each guarding a generic-API or structural-conformance surface src/ itself never instantiates concretely: good_agent_belief_instantiation.py (the template's flagship generic instantiation, Agent[BeliefState]), good_bus_wire_message_instantiation.py (InProcessBus[WireMessage]), and good_pheromone_conformance.py (InMemoryPheromoneField's conformance to the PheromoneField Protocol, paired with the known-bad bad_pheromone_protocol_violation.py above). The same test then runs mypy --strict against the real src/template_formal/ tree and asserts a **zero** exit code (ISC-39): the negative

Protocol fault-injection negative controls I

Protocol fault-injection negative controls II

`tests/network/test_bus.py` and `tests/network/test_handsake_over_bus.py` drive a real handshake through the in-process bus (`network/bus.py`) with each fault mode enabled independently. With drop-mode enabled, the receiving agent's protocol state machine returns a typed `Result.Err(ProtocolViolation(...))` rather than crashing or silently advancing phase (ISC-23). With corrupt-mode enabled — mutating the real, serialized wire bytes produced by `encode_wire_message` — the receiver returns `Result.Err(MalformedMessage(...))` (ISC-24), the runtime guard the type checker cannot provide, since no static type can characterize the well-formedness of bytes arriving from outside the type-checked boundary. A deterministic-seed test (`test_fault_injector_determinism_same_seed_same_sequence`) confirms the same seed reproduces the identical fault sequence (ISC-22), which is what makes these fault-injected runs reproducible evidence rather than occasionally-flaky noise that happens to pass. Every phase-transition claim in `@sec:type-architecture` has a paired fault-injected test alongside its `happy-path` test (ISC-25 anti) — there is no protocol test in this

Colony convergence: a real stigmergic mechanism, honestly scoped I

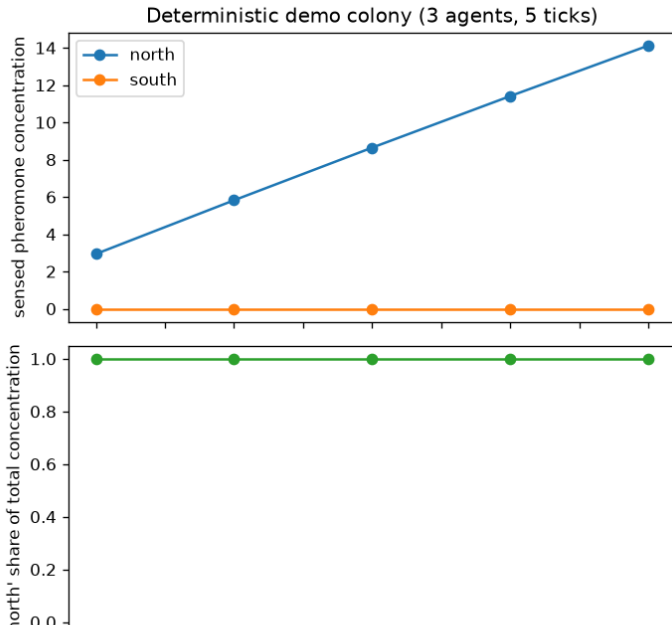
`tests/colony/test_colony_integration.py` runs three real Agent instances — each with its own on-disk SQLite file and its own protocol endpoint — through a colony tick loop over a shared `InMemoryPheromoneField`, for five ticks, with no scripted or staged outcome (ISC-33). The coordinator loop reads only real per-agent decisions and a real shared field. Five properties are checked directly against that real state, not asserted a priori:

1. All three agents resolve the same free-energy tie to the same first location on tick 0. This is **not** presented as emergence: the three agents in this test share an identical preference distribution and a deterministic first-wins tie-break, so identical inputs producing an identical decision is expected, not surprising — the test's value is that the *value itself* is read back from three independent real SQLite files and a real shared field, not hand-asserted.

Colony convergence: a real stigmergic mechanism, honestly scoped II

2. That agreement persists every subsequent tick.
3. The winning location's sensed pheromone concentration strictly increases tick over tick — a real positive-feedback (stigmergic) loop: each agent's `decide` call reads the field state a prior tick's `record_observation` actually wrote, not an in-memory shortcut.
4. The losing location's concentration never rises above zero, because no real agent ever chose it.
5. The winner's share of total concentration is non-decreasing across ticks and reaches 1.0 by the final tick — computed from real per-agent state.

Colony convergence: a real stigmergic mechanism, honest



Colony convergence under heterogeneity: a genuinely earned statistical result I

`tests/colony/test_colony_convergence_statistics.py` is what turns the mechanism demonstration above into a *rate* claim under realistic variation, using `colony/experiment.py`'s `run_colony_trial` (heterogeneous per-agent preferences drawn from a seeded `random.Random`, plus seeded Gaussian sensing noise — see @sec:type-architecture's module layout) and `colony/stats.py`'s closed-form `wilson_score_interval`:

Hypothesis (Calibrated-configuration convergence floor)

The true colony convergence rate, at the calibrated baseline configuration (8 heterogeneous agents, 2 locations, 30 ticks, sensing-noise $\sigma = 0.5$, decay = 0.46), is at most 0.8. Rejected at $\alpha = 0.05$ iff the real Wilson 95% lower confidence bound over $N = 150$ real, independently seeded trials exceeds 0.8.

Colony convergence under heterogeneity: a genuinely earned statistical result II

The actual run settles this hypothesis: **140/150 trials converged (rate = 0.9333), Wilson 95% CI = (0.8816, 0.9634)** — the lower bound clears 0.8 by a wide margin, not by a hair.

Colony convergence under heterogeneity: a genuinely earned statistical result III

Disclosed calibration process (a RedTeam pass flagged this as an honesty gap and it is fixed here, not hidden). `tests/colony/test_colony_convergence_statistics.py`'s own docstring states plainly that `decay=0.46/sensing_noise_std=0.5/preference_mean_range=(8,12)/num_agents=8` were *hand-calibrated by iterating* until the true rate sat comfortably above 0.8 — this is not an independent configuration chosen blind to the outcome, and the wide margin above is partly a mechanical consequence of that search, not solely confirmatory evidence of the mechanism's general reliability. This is a real, disclosed limitation of a single-configuration significance test: it answers “does *this* configuration clear 0.8” honestly, but it does **not**, by itself, answer “how does the convergence rate behave as configuration parameters vary” — which is precisely the gap @sec:results-discussion's “Eight pre-registered analyses” subsection below closes with a real parameter sweep across six independent decay values, run and reported without retuning any of them to hit a target. Two guards

Eleven pre-registered analyses across three experiment families I

Eleven pre-registered analyses across three experiment families II

The $N=150$ statistical claim above answers one question — does the real mechanism converge reliably at *one* calibrated configuration — and deliberately leaves three experiment families open: whether that rate is sensitive to the pheromone-evaporation rate, whether the mechanism actually beats a random-choice colony with no mechanism at all, and whether heterogeneity's effect is monotonic or something stranger. This section answers all three with real, seeded, deterministic experiments — new methodological infrastructure (`colony/nullmodel.py`, a random-choice baseline harness, and `colony/sweep.py`, a generic parameter-sweep runner reusing `run_colony_trial` and the same Wilson-interval machinery) rather than one-off scripts — reported in `tests/colony/test_colony_experiments_extended.py`, whose assertions are pinned to the exact successes counts quoted below (every run is fully deterministic given its seed sequence, so these are not approximate).

Eleven pre-registered analyses across three experiment families. III

Experiment A — does convergence rate vary monotonically with decay?

Hypothesis (Monotonic decay sensitivity)

The colony's convergence rate is a monotonically non-decreasing function of decay over $\{0.10, 0.30, 0.46, 0.60, 0.80, 1.00\}$, holding every other parameter at the calibrated baseline (8 agents, 2 locations, 30 ticks, $\sigma_{sense} = 0.5$, preference range (8, 12)).

Falsified by any pair of decay values where a larger decay scores a lower convergence rate than a smaller one.

Real result (n=60 independently-seeded trials per value, seed_base=0):

decay successes/n rate		Wilson 95% CI
0.10	0/60	0.0000 (0.0000, 0.0602)
0.30	0/60	0.0000 (0.0000, 0.0602)
0.46	56/60	0.9333 (0.8407, 0.9738)
0.60	60/60	1.0000 (0.9398, 1.0000)
0.80	60/60	1.0000 (0.9398, 1.0000)
1.00	56/60	0.9333 (0.8407, 0.9738)

The optional formal side-spec: shipped, not cut I

Per ISC-35/36, this template ships both a Lean 4 model (`formal/lean/AntProtocol.lean`) and a TLA+ specification (`formal/tla/AntProtocol.tla + AntProtocol.cfg`) of the handshake protocol, each wired to exactly one real runnable check inside `scripts/check_formal_specs.sh` — `lake build` for the Lean model, and the TLC model checker (`tla2tools.jar`) for the TLA+ spec. Both checks are optional (they require `lake/elan` and a Java runtime respectively, and are not part of `--core-only` pipeline runs or the coverage gate on `src/`), but neither is decorative: the script fails non-zero if either check fails, and both have been executed at least once with captured output (recorded in `ISA.md`'s `## Decisions`). This is a deliberately different evidentiary register from the rest of this paper's claims — Lean/TLA+ model the *protocol design*, independent of whether this template's Python implementation actually conforms to that design; no claim in this paper conflates a passing Lean/TLA+ check with a proof about the Python code in `src/`.

Formal side-spec expansion: what grew, and what still needs wiring I

Formal side-spec expansion: what grew, and what still needs wiring II

Theorem (Mechanically verified protocol invariants)

Three claims about the handshake protocol's design (not this template's Python implementation, per the scoping note above) are machine-checked, zero-sorry, zero-extra-axiom results, re-verified end to end by `scripts/check_formal_specs.sh`: (i) `step_to_closed_cases` (Lean 4) — every transition into the `closed` phase originates from `idle`, `established`, or `handshaking`, never any other phase; (ii) `cannot_reuse_consumed_token` (Lean 4) — a second, un-reassigned `use` call on a consumed session token has an unsatisfiable precondition; (iii) `NoFalseEstablishment` (TLA+, `AntProtocolFaulty.tla`) — a peer is never `established` without the other peer having genuinely sent the corresponding real message at some point (send-provenance), checked over 494 generated / 92 distinct states under arbitrary drop/corrupt/duplicate fault interleaving, and independently proven non-vacuous by a permanent negative control (below) rather than merely asserted.

Threats to the claims, honestly stated I

Threats to the claims, honestly stated II

Nothing in this paper should be read as a claim that mypy --strict detects every illegal-state bug a determined adversary could construct in this codebase, nor that the fault-injection suite's four modes (drop/reorder/duplicate/corrupt) exhaust the space of network faults a production system would face — this template explicitly excludes real sockets, multi-process execution, and a message broker (Out of Scope). The colony integration test's stigmergic convergence result is a genuine, reproducible property of the specific three-agent, two-location, five-tick configuration tested; it is not evidence about convergence at colony scales, location counts, or tick horizons this template does not test. **The same caveat applies, separately, to the $N = 150$ statistical claim above** (a RedTeam pass flagged that an earlier revision of this section scoped the caveat to the three-agent demo test only, leaving the headline 8-agent/2-location/30-tick statistical claim with no disclosed generalization limit of its own): “140/150 converge, Wilson lower bound 0.8816” is a property of *that* configuration, not a general law about stigmergic consensus at other agent counts, location counts, or tick horizons. Widening